Strings in Java

String, StringBuffer, and StringBuilder

Introduction

- A String is an object in Java that represents a sequence of characters.
- String is immutable (unchangeable once created).
- Strings are objects of the String class in java.lang package.

Characteristics of String

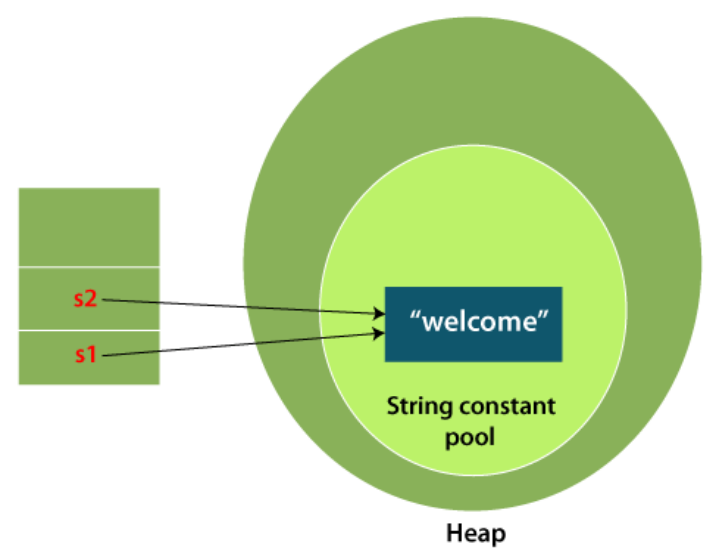
- Immutable: Once created, the string value cannot be modified.
- Stored in String Pool: Common strings are stored in the string pool for memory efficiency.
- Reference Type: String is an object, not a primitive type.
- Character Sequence: String holds Unicode characters (16-bit).

Different Ways to Declare Strings

String Literal

- This is the most common way to declare a string in Java.
- It creates a reference to the string literal “welcome", and Java stores this string in the string pool.
- If "Hello" already exists in the pool, the same reference is used.

```
String s1 = “welcome";
String s2 = “welcome";
```



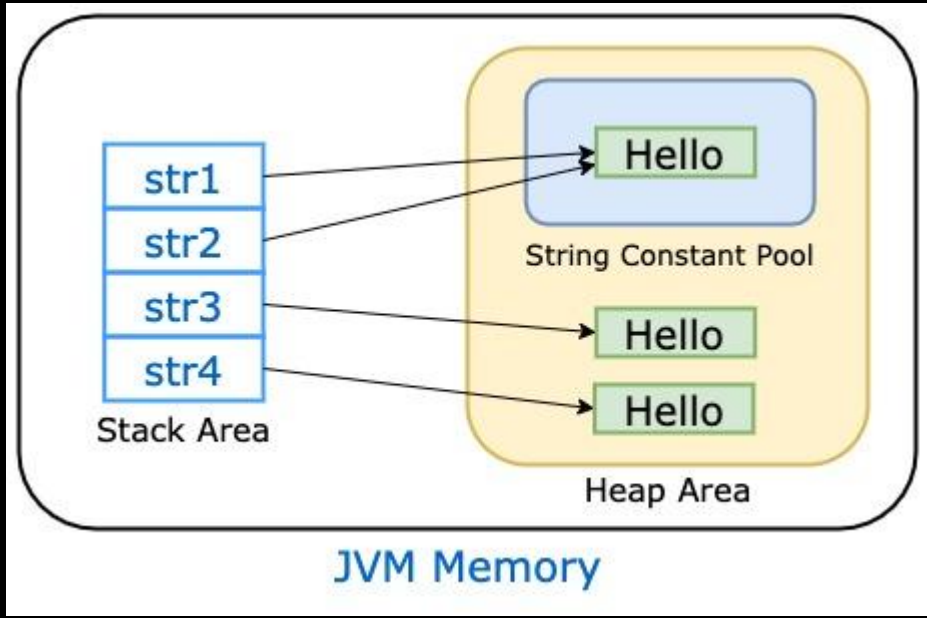
Different Ways to Declare Strings

Using new Keyword

- This creates a new String object in memory.
- So, the new keyword bypasses the string pool and creates a new string in heap memory, even if the exists in the string pool.
- This is generally less efficient.

```
String str = new String("Hello");
```

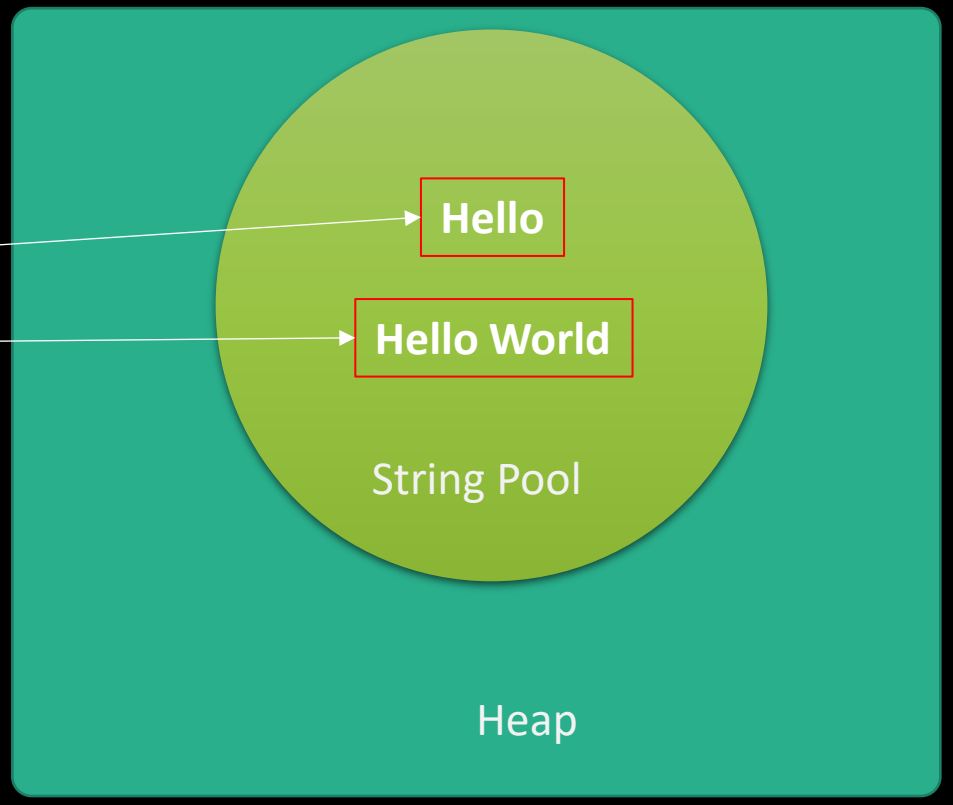
```
String str1 = “Hello”;
String str2 = “Hello”;
String str3 = new String(“Hello”);
String str4 = new String(“Hello”);
```



Immutable String

- In Java, String objects are immutable.
- Immutable simply means unmodifiable or unchangeable.

```
class Main {
  public static void main(String[] args) {
    String s1 = "Hello";
    s1 = s1+" World";
    System.out.println(s1);
}}
```



Why is String immutable?

- **Security:** Immutability ensures that string values cannot be changed, making them safe to use in situations where multiple threads might access the same string.

- **Performance:** Since strings are immutable, Java uses a string pool (interned strings) to reuse strings efficiently, avoiding the creation of multiple instances for the same value.

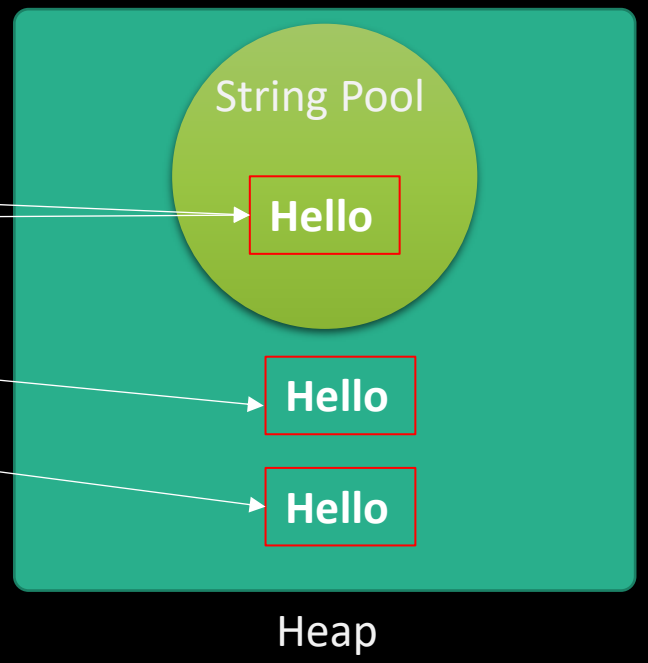
- Hashcode Caching: The immutability of strings makes it possible to cache their hashcode, improving the performance of operations like hashCode() and equals().

String Comparison (== vs equals())

- == compares references (memory location)
- equals() compares actual content (character sequence).

```
String str1 = "Hello";
String str2 = "Hello";
String str3 = new String("Hello");
String str4 = new String("Hello");

System.out.println(str1 == str2);  // true (same reference)
System.out.println(str1 == str3);  // false (different references)
System.out.println(str1.equals(str3));  // true (same content)
System.out.println(str3.equals(str4));  // true (same content)
```



Useful String Methods

- public int length() - Returns the length of the string.
- public char charAt(int index) - Returns the character at a specified index.
- public String substring(int beginIndex, int endIndex) - Extracts a substring.
- public String toUpperCase() - Converts the string to uppercase.
- public String toLowerCase() - Converts the string to lowercase.
- public String trim() - Removes leading and trailing whitespace.
- public String replace(char oldChar, char newChar) - Replaces a character or substring.
- int indexOf(String str) - Finds the first occurrence
- int lastIndexOf(String str) - Finds the last occurrence

Useful String Methods

- public boolean startsWith(String prefix) – checks whether the String starts with specified string.
- public boolean endsWith(String suffix) - checks whether the String ends with specified string.
- public boolean equals(Object anObject) - compares two strings, and returns true if the strings are equal, and false if not.
- public boolean equalsIgnoreCase(String anotherString) - Compares two strings for equality ignoring case.
- public int compareTo(String anotherString) - compares two strings lexicographically. returns 0 if equal, <0 if current string is less, and >1 if current string is greater.
- public String[] split(String regex) - Returns the index of the first occurrence of the specified substring. -1 if not found.

StringBuffer

- StringBuffer is used for mutable strings.
- It is thread-safe, meaning multiple threads can use it safely.
- Synchronized methods ensure thread safety but may slow down performance in single-threaded environments.

```
StringBuffer str1 = new StringBuffer("Hello");
str1.append(" World");  // Adds " World" to the string
System.out.println(str1);  // Output: "Hello World"
```

Important Methods of StringBuffer

- StringBuffer append(String str) - Adds data to the end of the current string.
- StringBuffer insert(int offset, String str) - Inserts data at a specified position.
- StringBuffer delete(int start, int end) - Removes characters from start index to end index (end index exclusive).
- StringBuffer reverse() - Reverses the content of the string.

StringBuilder

- StringBuilder is also used for mutable strings.
- Not thread-safe, but offers better performance in single-threaded environments.

```
StringBuilder builder = new StringBuilder("Hello");
builder.append(" World");
System.out.println(builder);  // Output: "Hello World"
```

Note: In Java, **thread-safe** describes code, classes, or methods that behave correctly when accessed by multiple threads simultaneously.

Important Methods of StringBuilder

- append() - Adds text to the end of the string.
- insert() - Inserts text at a specific position.
- delete() - Removes text from the string.
- reverse() - Reverses the content of the string.

StringBuffer vs StringBuilder

Feature	StringBuffer	StringBuilder
Thread Safety	Thread-safe (synchronized)	Not thread-safe
Performance	Slower (due to synchronization overhead)	Faster (no synchronization)
Use Case	Used in multi-threaded environments	Used in single-threaded or performance-critical applications